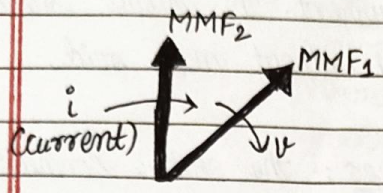
High Speed shaft:-
Spins at high speed driving the generator to produce electricity.



Recap

Wind Energy Conversion (WEC)

① Components

② classification → fixed speed
→ variable speed

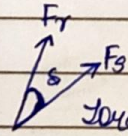
1. Fixed Speed (Synchronous Generator Schemes)

a) Constant frequency:-

b) Standalone scheme:-

c) Grid connected synchronous schemes

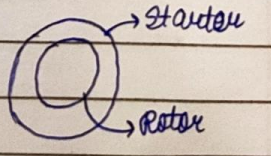
2. Variable Speed (Induction Generator Schemes)



Torque = $F_s F_r \sin\delta$

F_r → independent field

a) Grid connected
b) Self excited
c) Doubly-fed induction generator
d) AC-DC-AC



Motor (I-ϕ induction motor)

Grid connected:- The stator connects straight to the utility grid. The grid supplies the reactive power needed for the excitation. External capacitor bank and soft starters support voltage levels.

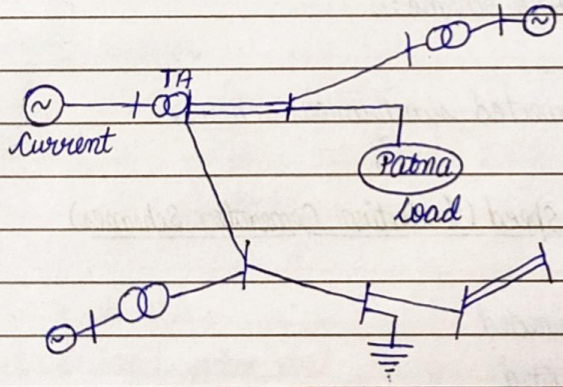
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Self excited Induction generator schemes :- External capacitor banks connects across the stator terminal to supply the reactive magnitizing current. this allows standalone operation in remote area without main grid.

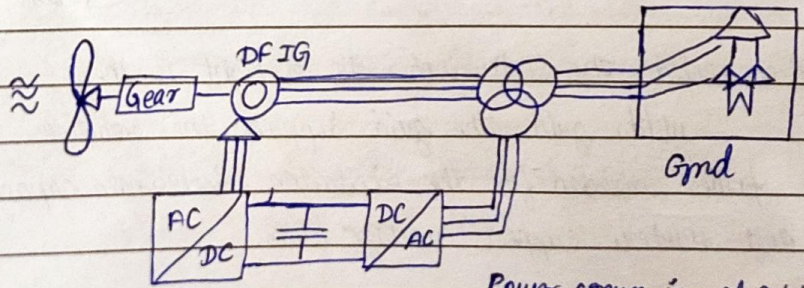
Doubly-fed Induction generator schemes :- The stator connects directly to the grid, while the rotor connects through a partial-scale power converter. This variable-speed scheme maximizes power capture across changing turbine speed.

EV { AC-DC-AC line Double Output schemes :- Rectifies and inverts power or uses slip-ring commutator system to handle variable rotor frequencies and feed power back into the system.

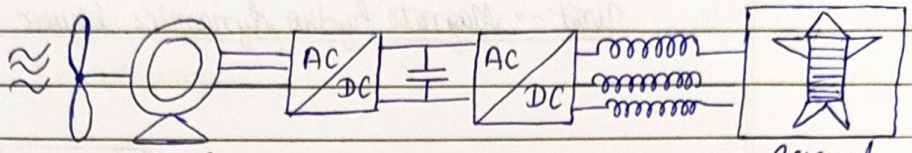
* Wind Energy conversion system connected to power grid



Complex network



Power conversion of Wind



Advantages :-

- Eco-friendly
- Inexhaustible
- Sustainable Growth

Limitations :-

- Intermittency
- High initial cost
- Location constraints

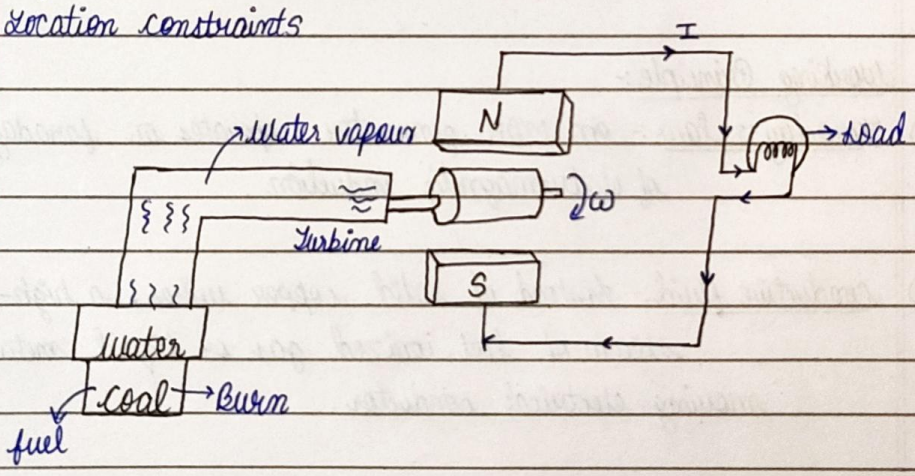


Fig :- coal-fired power plant

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Unit → Magneto hydro dynamics power generation

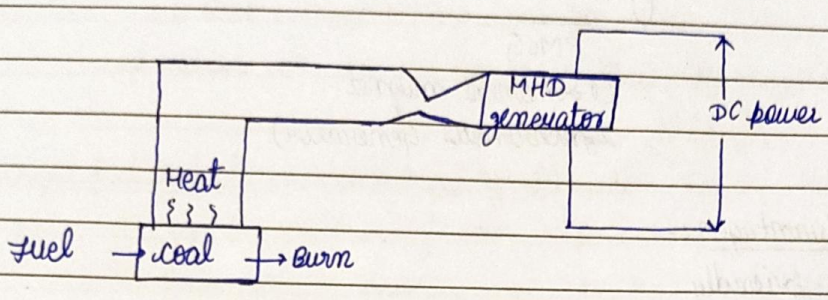


Fig:- Coal fired power plant

→ is an advanced direct energy conversion method that transforms thermal and kinetic energy from a moving electrically conductive fluid directly into electricity without moving parts.
(produce DC)

Working Principle:-

a) Faraday's law :- An MDH generator operates on faraday's law of electromagnetic induction.

b) Conductive fluid:- Instead of solid copper wires, a high-velocity steam of hot, ionized gas or liquid metal acts as the moving electrical conductor.

c) Magnetic field :- The conducive fluid is forced through a strong magnetic field inside a channel or duct.

d) Current generation:- The movement induces a voltage perpendicular to both the fluid flow and the magnetic field, and electrodes on the channel walls collect the resulting direct current (DC power).

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→ Types of system
→ Advantage & disadvantages